

# Deep Learning for teens

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# TF Input Pipeline

- ▶ A TensorFlow Input Pipeline is just a way to get your data ready for your model. It loads your data step-by-step so your model never has to wait.
- ▶ It's a simple way of saying that the TF Input Pipeline is like a pipeline that prepares data—doing things like scaling, batching, and other preprocessing steps—so that your model can read it easily.
- ▶ <https://www.kaggle.com/code/hamidsadeghi1/mnist-input-pipeline-demo>

# Optimize tensorflow pipeline

- ▶ Absolutely! Optimizing the TensorFlow pipeline is really about using the right tools to speed up data loading and processing.
- ▶ **Step 1: Cache the Dataset**
  - ▶ `dataset = dataset.cache()` # This will cache the dataset in memory after the first load
- ▶ **Step 2: Prefetch the Data**
  - ▶ `dataset = dataset.prefetch(buffer_size=tf.data.AUTOTUNE)` # This will allow later elements to be prepared while the model is training on current elements
- ▶ Prefetching is essentially about getting the next batch of data ready in the background while your model is still working on the current batch. It means the pipeline is always one step ahead, so your model doesn't have to pause and wait for new data to be loaded. So it's just a way to keep things running smoothly!

# Homework

- ▶ Use the KMNIST dataset to do the same steps I have done in the coding part.
- ▶ KMNIST on Kaggle:
  - ▶ <https://www.kaggle.com/datasets/victoriaacuna/kmnist>